LESSON 3.9

Equivalent Ratios

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find and check equivalent ratios using multiplication and division.

WHY THIS MATTERS Chef Montoya is preparing her famous tomato soup for a school banquet. Her original recipe serves 8 people, but tonight she needs to serve 32 guests. She needs your help to scale the recipe without changing the flavor!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Equivalent Ratios *Spanish: Razones equivalentes*	Ratios that compare quantities in the same relationship.	2:3 = 4:6 because both parts were multiplied by 2.
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	3 red apples : 5 green apples
Rate *Spanish: Tasa*	A ratio comparing two amounts with different units, like miles per hour.	60 miles per 1 hour
Proportion *Spanish: Proporción*	A math sentence saying two ratios are equal.	2/3 = 4/6
Simplify *Spanish: Simplificar*	To write a ratio with smaller numbers that still compares the same way. 6:8 and 3:4 are the same ratio.	12:18 → divide both by 6 → 2:3
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that shows a pattern.	Cups of juice: 2, 4, 6 | Cups of water: 3, 6, 9

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Chef Montoya's soup serves 8 people, but she needs to serve 32.

1. I find the scale factor: 32 divided by 8 = 4.
2. The recipe uses 3 cups of tomatoes to 2 cups of broth, a 3:2 ratio.
3. I multiply both by 4: 3 x 4 = 12 cups of tomatoes, and 2 x 4 = 8 cups of broth.
4. So 3:2 and 12:8 are equivalent ratios, and the flavor stays the same.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A ratio is 2:5. We want to make it bigger using a scale factor of 3.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I multiplied _____ and _____ by the same number, _____, so the ratio _____ is equivalent to 3:2.

Word bank Equivalent Ratios | Ratio | Rate | Proportion | Simplify | Ratio table | multiply | same number | equivalent | scale | tomatoes | broth

WATCH OUT!

A common mistake

A common mistake in Equivalent Ratios is ADDING the scale factor to one part instead of MULTIPLYING both parts by it. For example, scaling 2:5 by a factor of 4 should give 8:20 (2×4 and 5×4)

LESSON 3.9

Equivalent Ratios

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each pair of ratios: are they equivalent or not equivalent?

Groups: Equivalent Ratios | NOT Equivalent

_____ 2:4 and 3:6

_____ 5:10 and 1:2

_____ 3:7 and 6:12

_____ 4:9 and 8:16

HINT Two ratios are equivalent if one is the other with BOTH parts scaled by the same multiplier.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which ratio is equivalent to 2:5?

(A) 2:10 (B) 3:6

(C) 4:10 (D) 5:2

HINT Equivalent ratios come from multiplying BOTH parts of 2:5 by the same number.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A recipe uses 4 eggs for every 6 cups of flour. Which ratio is equivalent?

(A) 2:3
(B) 4:3
(C) 6:4
(D) 8:10

HINT You can scale a ratio DOWN too — divide both parts by the same number.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which ratio is NOT equivalent to 6:9?

(A) 2:3
(B) 3:5
(C) 12:18
(D) 18:27

HINT First simplify 6:9 to its simplest form — that becomes your measuring stick.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the simplest form of the ratio 15:25?

(A) 1:5
(B) 3:5
(C) 5:3
(D) 15:25

HINT Simplest form means both numbers have been divided by their greatest common factor.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake — Lemonade Ratio

1. **Set up the ratio** 2 cups lemon : 5 cups water = 6 cups lemon : ? cups water
2. **Find the multiplier** 6 ÷ 2 = 3
3. **Apply to the water** 5 + 3 = 8
4. **Answer** 6 cups lemon : 8 cups water

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Look at Step 3. The lemon was multiplied by 3, so what should you do to the water?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you find the missing values in Chef Montoya's table? How can you PROVE two of your rows are equivalent ratios?

> **SENTENCE STARTER** I multiplied ___ and ___ by the same number, ___, so the ratio ___ is equivalent to 3:2.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.9

Equivalent Ratios

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort these ratios — which are equivalent to 3:4 and which are not?

Groups: Equivalent to 3:4 | NOT Equivalent to 3:4

_____ 6:8

_____ 9:12

_____ 15:20

_____ 3:5

_____ 4:3

_____ 6:12

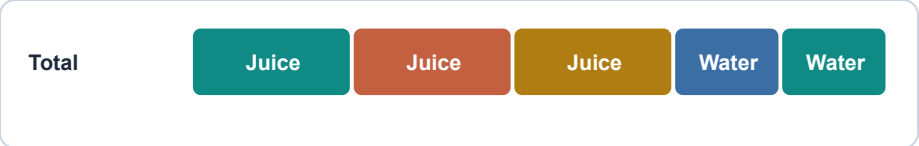
2 **Complete the table. Show how you found each value.**

Find the missing value that makes each pair of ratios equivalent.

Ratio 1	Ratio 2	Missing Value
3:7	9:?	
5:8	?:24	
4:6	?:3	

SHOW YOUR WORK

3 Use the model to solve. Show your work.



Each part = 25 ÷ 5 = 5 cups. Juice = 3 × 5 = 15 cups. Water = 2 × 5 = 10 cups.

SHOW YOUR WORK

Answer: _____

PART 2 Apply it

real-world problems

4 Circle the letter of the best answer. Show how you know.

Marcus earns $45 for every 3 hours of work. How much does he earn in 8 hours?

(A) $90
(B) $105
(C) $120
(D) $135

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake — Simplifying a Ratio

1. **Set up the ratio** A student wants to write 12:18 in simplest form.
2. **Find the common factor** Both 12 and 18 can be divided by 6.
3. **Divide each part** 12 ÷ 6 = 2 and 18 ÷ 3 = 6, so they wrote 2:6.
4. **Answer** 12:18 simplifies to 2:6

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you find the missing values in Chef Montoya's table? How can you PROVE two of your rows are equivalent ratios?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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Equivalent Ratios

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

A student claims these ratios are all equivalent. Check each one and mark whether it simplifies to 2:5 or not.

Ratio	Simplified Form	Equivalent to 2:5?
4:10	2:5	Yes
6:15		
8:25		
10:25		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Marcus's Mistake

1. **Set up the ratio** 3 cups : 5 cups = ? : 20 cups
2. **Find the multiplier** 20 ÷ 5 = 4
3. **Apply to both parts** 3 + 4 = 7
4. **Answer** 7 cups : 20 cups

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A store sells 3 pens for $2. Another store sells 9 pens for $6. Are these equivalent ratios? If so, prove it using at least two different methods (simplifying, scaling, or cross-multiplying).

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How did you find the missing values in Chef Montoya's table? How can you PROVE two of your rows are equivalent ratios?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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